

ECO 529: Macro-Finance: Tools, Models, and Ideas

Lecture 6: Equilibrium

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Goals

- ▶ Introduce continuous time competitive equilibrium (so called “Mean Field Game”)
 - ▶ Brings together optimal control and evolution of distribution
 - ▶ Characterised by coupled HJBE and KFE or by “master equation”
- ▶ Examples: Aiyagari-Bewley-Hugget model and Krusell-Smith Model. Why these?
 - ▶ Because they are building block models for many papers (e.g. HANK).
 - ▶ These also have simple market clearing conditions.
- ▶ Compare rational expectations, self confirming, and misspecified belief equilibria.

References

- ▶ Original uninsurable idiosyncratic risk models: Bewley (1980 & 1983), Huggett (1993), Aiyagari (1994) and Krusell and Smith, (1998)
- ▶ Continuous time formulations:
 - ▶ Achdou, Han, Lasry, Lions, Moll. “Income and Wealth Distribution in Macroeconomics: A Continuous-Time Approach” (2017), and
 - ▶ Gu, Lauriere, Merkel, Payne. “Deep Learning Solutions to Master Equations for Continuous Time Heterogeneous Agent Macroeconomic Models” (2023).
- ▶ Mean field games: Achdou, Han, Lasry, Lions, Moll (2017), Gomes et al. (2015), Lacker (2015), Carmona and Delarue (2018)

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Baseline Macroeconomic Model With Simple Asset Pricing (Krusell-Smith (1998))

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Aiyagari (1994) and Krusell & Smith (1998)

- ▶ Benchmark models in “quantitative” macroeconomics.
- ▶ Aiyagari-Bewley-Huggett (ABH) (1994,1980,1993): model of precautionary saving:
 - ▶ Households get idiosyncratic shocks to labour hours (i.e. employment $\leftrightarrow$ unemployment).
 - ▶ Frictions: (i) idiosyncratic labour risk is uninsurable, (ii) exogenous borrowing limit.
 - ▶ Households want to borrow when labour hours are low so they can smooth consumption but their borrowing limits prevent it.
 - ▶ So, instead, households save assets to sell when labour hours are low
- ▶ Krusell & Smith (1998): introduces aggregate TFP shocks into ABH model.

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Setting, Agents and Preferences

- ▶ Continuous time, infinite horizon economy with one consumption good.
- ▶ Population: continuum of agents, denoted by $I = [0, 1]$.
- ▶ Preferences: each agent, $i \in I$, ranks outcomes according to:

$$\mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} u(c_t^i) dt \right],$$

- ▶ c_t^i is agent i 's consumption at time, t , and
 - ▶ $u(c) = c^{1-\gamma}/(1-\gamma)$ is the flow utility from consuming c
- ▶ Household labour endowment, $n_t \in \{n_1, n_2\}$, follows continuous time Markov chain:
 - ▶ Process jumps from n_1 to n_2 according Poisson process with intensity rate λ_1
 - ▶ Process jumps from n_2 to n_1 according Poisson process with intensity rate λ_2

Production Technology

- Representative firm production technology:

$$Y_t = e^{Z_t} F(K_t, L_t)$$

- K_t, L_t : capital and labor hired at time t
- F : CRS production function (E.g. $F(K, L) = K_t^\alpha L_t^{1-\alpha}$)
- Aiyagari (1994): $Z_t = Z$ is constant so there is no aggregate risk.
- Krussel-Smith (1998): Z_t : aggregate productivity follows a Ornstein-Uhlenbeck process:

$$dZ_t = \eta(\bar{Z} - Z_t)dt + \sigma dB_t^0$$

B_t^0 is a common Brownian motion process; it generates filtration $\mathcal{F}_t^0$.

- Consumption goods can be converted into capital at 1 – 1 ratio (and vice-versa).
- Capital depreciates at rate δ .

Markets

- ▶ Incomplete financial markets:
 - ▶ Agents cannot insure idiosyncratic income risk.
 - ▶ Instead, agents can “save and borrow claims on capital”
 - ▶ Let a_t denote agent’s claim on capital stock.
 - ▶ Face exogenous borrowing constraint: $a_t \geq \underline{a} > -\infty$
- ▶ Competitive capital market with rental rate r_t . Clearing implies: $\int_I a_t^i di = K_t$
- ▶ Competitive labour market with wage rate w_t . Clearing implies: $\int_I n_t^i di = L_t$
- ▶ Goods markets (clears by Walras’s Law).

Q & A

- ▶ **Q.** What does it mean to have a negative claim on capital?
 - ▶ E.g. there is a frictionless financial intermediary
 - ▶ Households save into and borrow from the intermediary
 - ▶ Intermediary uses the savings to create capital and rent it to firm

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Firm Problem

- Representative firm solves:

$$\max_{K_t, L_t} \{e^{Z_t} F(K_t, L_t) - (r_t + \delta)K_t - w_t L_t\}$$

- First order conditions:

$$\begin{aligned}(K_t) : \quad r_t &= e^{Z_t} \partial_K F(K_t, L_t) - \delta, \\(L_t) : \quad w_t &= e^{Z_t} \partial_L F(K_t, L_t)\end{aligned}$$

Household Problem

- ▶ Household i has **idiosyncratic state** $x_t^i = (a_t^i, n_t^i)$, where a_t^i is wealth, n_t^i is labor.
- ▶ Given belief about **price process**, household chooses **consumption** $c = \{c_t^i\}_{t \geq 0}$ to solve:

$$\begin{aligned} \max_{\{c_t^i\}_{t \geq 0}} \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} u(c_t^i) dt \right] \\ \text{s.t.} \quad da_t^i = (\tilde{w}_t n_t^i + \tilde{r}_t a_t^i - c_t^i) dt =: s(c_t^i, a_t^i, n_t^i, \tilde{w}_t, \tilde{r}_t) =: \mu_t^a dt, \quad a_t^i \geq \underline{a} \\ n_t^i \in \{n_1, n_2\}, \text{ switches at idiosyncratic Poisson rate } \lambda(l_t^i) \end{aligned} \quad (1)$$

- ▶ $(\tilde{r}, \tilde{w}) = \{\tilde{r}_t, \tilde{w}_t\}_{t \geq 0}$ are agent beliefs about prices processes,
- ▶ $\underline{a}$: **borrowing limit**.
- ▶ $G_t = \mathcal{L}(x_t^i | \mathcal{F}_t^0)$ & g_t are **population distribution** & **density** of $(x_t^i)_{i \in I}$, given **history** $\mathcal{F}_t^0$
 - ▶ Non degenerate because households get uninsurable idiosyncratic labor shocks.

Kolmogorov Forward Equation

- ▶ Given stochastic processes $\{c, \tilde{r}, \tilde{w}\}$, at time $t > 0$, on $(\underline{a}, \infty)$, the **density** $g_t(a, n)$ satisfies the **KFE**:

$$\partial_t g_t(a, n) = -\partial_a((\tilde{w}_t n + \tilde{r}_t a - c_t)g_t) + \underbrace{\lambda(\check{n})g_t(a, \check{n})}_{\text{Entry from } \check{n} \rightarrow n} - \underbrace{\lambda(n)g_t(a, n)}_{\text{Exit from } n \rightarrow \check{n}}$$

- ▶ Observations:
 - ▶ Why does the KFE not have dW_t^0 ?
 - ▶ What changes to the environment would lead to a dW_t^0 appearing?
 - ▶ $g_t(a, \tilde{y})$ is still a stochastic process in equilibrium since r and w will be stochastic processes adapted to history of common shocks $\mathcal{F}_t^0 = \sigma(\{W_s^0 : 0 \leq s \leq t\})$.

Rational Expectations Equilibrium

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief that price process is $\tilde{q}$, household consumption process, c_t^i , solves (1),
2. Given belief that price process is $\tilde{q}$, firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions**:

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. Agent beliefs about the price process are **consistent**: $\tilde{q} = q$

Rational Expectations Equilibrium (Combining Equations For Prices)

Definition: Given an initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, q_t := [r_t, w_t], z_t : t \geq 0, i \in I\}$, s.t.:

1. Given belief that price process is $\tilde{q}$, household consumption process, c_t^i , solves (1),
2. The price vector $q_t = [r_t, w_t]$ satisfies:

$$q_t = \begin{bmatrix} r_t \\ w_t \end{bmatrix} = \begin{bmatrix} e^{z_t} \partial_K F(K_t, L) - \delta \\ e^{z_t} \partial_L F(K_t, L) \end{bmatrix} =: Q(z_t, g_t), \text{ where } K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da$$

3. Agent beliefs about the price process are **consistent**: $\tilde{q} = q$

Having explicit expression for prices in terms of (z_t, g_t) makes problem very tractable.
(We consider more complicated cases later.)

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Partially Recursive Representation of Equilibrium

- Aggregate states: t , individual states: (a, n) , stoch. household value fn: $V_t(a, n)$.
- Given belief $(\tilde{r}, \tilde{w}) = \{(\tilde{r}_t, \tilde{w}_t)\}_{t \geq 0}$, household (a, n) chooses c to solve stoch. HJBE:

$$dV_t(a, n) = \max_c \left\{ -\rho V_t(a, n) + u(c) + \partial_a V_t(a, n)(\tilde{w}_t n + \tilde{r}_t a - c) + \lambda(n) (V_t(a, \check{n}) - V_t(a, n)) \right\} dt + \Theta_t(a, n) dW_t^0, \quad \text{s.t.} \quad \text{BC: } \frac{\partial V}{\partial a} \Big|_{a=\underline{a}} \geq u'(\tilde{w}_t \underline{n} + \tilde{r}_t \underline{a})$$

where $\Theta_t(a, n)$ is unknown and $\check{n}$ is complement of n .

- For optimal policy, $c_t^*(a, n; \tilde{r}, \tilde{w})$, for z_t , population density, g , evolves by KFE:

$$dg_t(a, n) = \underbrace{[-\partial_a[(\tilde{w}_t n + \tilde{r}_t a - c_t^*)g_t(a, n)] - \lambda(n)g_t(a, n) + \lambda(\check{n})g_t(a, \check{n})]dt}_{=:\mu_t^g(c^*, a, n; \tilde{r}, \tilde{w})}$$

- Agent beliefs about price process are consistent with market clearing: $\tilde{r} = r$, $\tilde{w} = w$.

Recursive Representation of Equilibrium

- Aggregate states: (z, g) , individual states: (a, n) , household value fn: $V(a, n, z, g)$.
- Given belief $dg_t = \tilde{\mu}^g(z, g)dt$, household at state (a, n) chooses c to solve **HJBE**:

$$0 = \max_c \left\{ -\rho V(a, n, z, g) + u(c) + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c) \right. \\ \left. + \lambda(n) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g)\mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \right. \\ \left. + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n, z, g(y, n_j)) \tilde{\mu}^g(y, n_j, z, g) dy \right\}, \quad \text{s.t.} \quad \text{BC: } \frac{\partial V}{\partial a} \big|_{a=\underline{a}} \geq u'(wn + r\underline{a})$$

where $\frac{\partial V}{\partial g}(a, n, z, g(\cdot))$ is Frechet derivative w.r.t. g and $\check{n}$ is complement of n .

- For optimal policy, $c^*(a, n, z, g; \tilde{\mu}^g)$, for z_t , population density, g , evolves by **KFE**:

$$dg_t(a, n) = \underbrace{[-\partial_a [(w(z, g)n + r(z, g)a - c^*)g_t(a, n)] - \lambda(n)g_t(a, n) + \lambda(\check{n})g_t(a, \check{n})]dt}_{=:\mu^g(a, n, z, g; \tilde{\mu}^g)}$$

- In equilibrium, beliefs are consistent $\tilde{\mu}^g = \mu^g$.

Technical Aside: Hamilton-Jacobi-Bellman Equation

- The HJBE is:

$$\begin{aligned} 0 = \max_c \bigg\{ & -\rho V(a, n, z, g) + u(c) + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c) \\ & + \lambda(n) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g)\mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \\ & + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n, z, g(y, n_j)) \tilde{\mu}^g(y, n_j, z, g) dy \bigg\}, \quad s.t. \quad \text{BC: } \frac{\partial V}{\partial a} \big|_{a=\underline{a}} \geq u'(wn + r\underline{a}) \end{aligned}$$

- Recall technical aside: No “classical solution” because the value function is not differentiable when the borrowing constraint binds. Instead, need to show that a “viscosity” solution exists.
- Recall results (see [Achdou et. al. \(2018\)](#)):
 - Borrowing constraint never binds in the interior of state space
 - Instead, borrowing constraint appears as a boundary inequality

Technical Aside: Kolmogorov Forward Equation

- ▶ On $(\underline{a}, \infty)$, the density $g_t(a, y)$ satisfies the KFE:

$$dg_t(a, n) = [-\partial_a[(w(z, g)n + r(z, g)a - c^*)g_t(a, n)] - \underbrace{\lambda(n)g_t(a, n)}_{\text{Exit from } n \rightarrow \check{n}} + \underbrace{\lambda(\check{n})g_t(a, \check{n})}_{\text{Entry from } \check{n} \rightarrow n}]dt$$

- ▶ Technical asides:
 - ▶ Low income agents get “stuck” at borrowing constraint $\Rightarrow$ mass point in the density.
 - ▶ So KFE will not have “classical solution”. Instead, need to use a type of measure value solution (see [Achdou et. al. \(2018\)](#)).
 - ▶ The KFE not have dB_t^0 . Why?
 - ▶ What changes to the environment would lead to a dB_t^0 appearing in KFE?

Inequality boundary condition causes technical problems $\Rightarrow$ better to soften it!

Recursive Representation of Equilibrium (Soft Borrowing Constraint)

- Aggregate states: (z, g) , individual states: (a, n) , household value fn: $V(a, n, z, g)$.
- Given a belief $dg_t = \tilde{\mu}_g(z_t, g_t)dt$, household at $x = (a, n)$ choose c to solve HJBE:

$$0 = \max_c \left\{ -\rho V(a, n, z, g) + u(c) - \mathbf{1}_{a_t \leq \underline{a}} \psi(a_t) + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c) \right. \\ \left. + \lambda(n)(V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g)\mu^z(z) + 0.5(\sigma^z)^2 \partial_{zz} V(a, n, z, g) \right. \\ \left. + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n, z, g) \tilde{\mu}^g(\cdot) dy \right\}, \quad s.t. \quad \text{BC: } \frac{\partial V}{\partial a} \Big|_{a=\underline{a}} \geq u'(\cdot), \quad \psi(a) = -\frac{1}{2} \kappa(a - \underline{a})^2$$

where $\check{n}$ is complement of n .

- For optimal policy, $c^*(a, n, z, g; \tilde{\mu}^g)$, for z_t , population density, g , evolves by KFE:

$$dg_t(a, n) = \underbrace{[-\partial_a[(w(z, g)n + r(z, g)a - c^*)g_t(a, n)] - \lambda(n)g_t(a, n) + \lambda(\check{n})g_t(a, \check{n})]}_{=:\mu_t^g(a, n, z, g; \tilde{\mu}^g)} dt$$

- In equilibrium, beliefs are consistent $\tilde{\mu}^g = \mu^g$.

“Master Equation” Representation of Equilibrium

- ▶ “Master equation” substitutes KFE, market clearing, & belief consistency into HJBE.
- ▶ Equilibrium value function $V(a, n, z, g)$ characterized by one PDE:

$$\begin{aligned} 0 = & -\rho V(a, n, z, g) + u(c^*(a, n, z, g)) + \mathbf{1}_{a_t \leq \underline{a}} \psi(a_t) \\ & + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c^*(a, n, z, g)) \\ & + \lambda(x) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g) \mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \\ & + \sum_{j \leq 2} \int_{\underline{a}}^{\infty} \frac{\partial V}{\partial g}(a, n_j, z, g(y, n_j)) \mu^g(c^*(y, n_j, z, g), y, n_j, z, g) dy =: \mathcal{L}V \end{aligned}$$

where the optimal control c^* is characterised by:

$$u'(c^*(a, n, z, g)) = \partial_a V(a, n, z, g).$$

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Three Notions of equilibrium

- ▶ During the 1980s and 1990s, there was a lot of interest in relaxing the rational expectations equilibrium.
- ▶ See [[Sargent, 2008](#)] and [[Sargent, 1999](#)]

Three notions of equilibrium

1. Rational expectations equilibrium
2. Self confirming belief equilibrium (chapter 7 in [[Sargent, 1999](#)])
3. Optimal misspecified belief equilibrium (chapter 6 in [[Sargent, 1999](#)])

1. Rational expectations equilibrium

- ▶ Agents have a “perceived law of motion” for the economic variables that is the true law of motion for the economy if agents follow their perceived law of motion (on and off the equilibrium path).
- ▶ In this sense, agents understand the law of motion even for data that would never be generated on the equilibrium path.

1. Rational Expectations Equilibrium in KS-98

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief that price process is $\tilde{q}$, household consumption process, c_t^i , solves (1),
2. Given belief that price process is $\tilde{q}$, firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions**:

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. Agent beliefs about the price process are **consistent**: $\tilde{q} = q$

2. Self confirming belief equilibrium

- ▶ Agents have a “perceived law of motion” for the equilibrium variables in the economy.
- ▶ In equilibrium, this law of motion is the best statistical fit to data generated by an economy where agents follow that perceived law of motion. In this sense, agents would never reject a misspecification test on their law of motion.
- ▶ However, they may have off-equilibrium beliefs that differ from rational expectations.
- ▶ So, this is like an “indoctrination” equilibrium: agents all believe the world follows particular rules, which influences their behavior and prevents the economy from generating data that could lead them to reject their beliefs.

2. Self Confirming Belief Equilibrium in KS-98

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief that price process is $\tilde{q}$, household consumption process, c_t^i , solves (1),
2. Given belief that price process is $\tilde{q}$, firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions**:

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. *Agent beliefs about the price process $\tilde{q}$ are the **best fit** based in any data generated from the model under agent belief $\tilde{q}$.*

2. Self Confirming Belief Equilibrium: A Simple Conceptual Example

- ▶ Suppose that we live in political regime A .
- ▶ Suppose that political regime B is actually a better regime but it has never been tried anywhere in the world and we all believe that regime A is better than regime B .
- ▶ Then always choosing political regime A is a self-confirming equilibrium:
 - ▶ Continuing with political regime A is optimal given the belief that A is better
 - ▶ And there is no data that is ever generated that could disprove this belief.
- ▶ However, choosing political regime A is not a rational expectations equilibrium:
 - ▶ Agents in rational expectations correctly believe that regime B is actually better.
 - ▶ So, continuing with regime A is not optimal.

3. Optimal Misspecified Belief Equilibrium: “Dogmatic Form”

- ▶ Agents have a “perceived law of motion” for the equilibrium variables in the economy that *satisfies a particular parametric form*.
- ▶ In equilibrium, the parameters in their perceived law of motion are the statistical best fit to the data generated by an economy where all agents follow that perceived law of motion.
- ▶ However, if agents were to perform a misspecification test on the parametric form for their perceived law of motion, they would typically reject it.

3. Optimal Misspecified Belief Equilibrium in KS-98 (Example)

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief that price process is $\tilde{q}$, household consumption process, c_t^i , solves (1),
2. Given belief that price process is $\tilde{q}$, firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions**:

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. *Agent beliefs about the price process $\tilde{q}$ (i) follow a parametric rule $\mu_q = \mu(K_t; \theta)$ and $\sigma_q = \sigma(K_t; \theta)$ and (ii) the parameters θ are the “best response” to data generated by the economy under $\mu_q = \mu(K_t; \theta)$ and $\sigma_q = \sigma(K_t; \theta)$.*

3. Optimal Misspecified Belief Equilibrium: “Learning”

- ▶ Agents have a “perceived law of motion” for the equilibrium variables in the economy.
- ▶ In equilibrium, the parameters in their perceived law of motion are the statistical best fit *subject to cognitive or physical updating constraints* to the data generated by an economy where all agents follow that perceived law of motion.
- ▶ However, if agents were to perform a misspecification test on the parametric form for their perceived law of motion, they would typically reject it.

3. Optimal Misspecified Belief Equilibrium in KS-98 (Example)

Definition: Given initial density g_0 , an **equilibrium** for this economy consists of a collection of $\mathcal{F}_t^0$ -adapted stochastic process, $\{c_t^i, g_t, z_t, q_t := [r_t, w_t] : t \geq 0, i \in I\}$, s.t.:

1. Given belief that price process is $\tilde{q}$, household consumption process, c_t^i , solves (1),
2. Given belief that price process is $\tilde{q}$, firm choose capital and labor optimally:

$$r_t = e^{z_t} \partial_K F(K_t, L) - \delta, \quad w_t = e^{z_t} \partial_L F(K_t, L)$$

3. The price vector $q_t = [r_t, w_t]$ satisfies **market clearing conditions**:

$$K_t = \sum_{j \in \{1,2\}} \int a g_t(a, n_j) da, \quad L = \sum_{j \in \{1,2\}} \int n_j g_t(a, n_j) da$$

4. *Agent beliefs about the price process $\tilde{q}$ follow an updating rule that is governed by cognitive or physical constraints.*

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Next Lectures: Solve for These Types of Equilibria

- ▶ Rational expectations equilibrium: train neural network to solve the master equation (e.g. Gu-Lauriere-Merkel-Payne (2023))
- ▶ Self confirming equilibrium: model based reinforcement learning (e.g. Han-Yang-E (2021))
- ▶ Optimal misspecified belief equilibrium: many approaches depending on the type of belief misspecification:
 - ▶ Original Krussel-Smith algorithm is a “dogmatic” misspecified belief equilibrium.
 - ▶ “Bounded” rationality models
 - ▶ Incomplete model/model free based reinforcement learning

Which equilibrium concepts are vulnerable to the Lucas critique?

First Approach: Rational Expectations

- ▶ Goal: solve Master equation numerically
- ▶ Problem: Master equation contains an infinite dimensional derivative.
- ▶ Solution: three main ingredients:
 1. High but finite dimensional approximation to distribution and Master equation:
 - (i). Replace continuum of agents by a **finite population** of agents, or
 - (ii). **Project** distribution onto a finite dimensional set of basis functions (e.g. indicator functions, eigenfunctions, Chebyshev polynomials ...).
 2. Parameterize V by neural network, and
 3. Train the parameters to minimize the (approximate) master equation residual.

References I



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Appendix

Aside: Infinite Dim. Calculus

- ▶ Let X be a Hilbert space with inner product $\langle \cdot, \cdot \rangle_X$ and let $Y \subset \mathbb{R}$. Define an inner product on $B(X, Y) = X^*$ by:

$$\langle f, g \rangle_{X^*} \equiv \int_X f(x)g(x)dx, \quad \forall f, g \in B(X, Y)$$

- ▶ Let F map a bounded linear function $f \in B(X, Y)$ into Y . That is:

$$F : B(X, Y) \rightarrow Y, \quad f \mapsto F(f)$$

- ▶ Let DF denote either the Frechet derivative. Then, for each $f \in B(X, Y)$ we have that:

$$DF(f) : B(X, Y) \rightarrow Y, \quad \eta \mapsto DF(f)\eta$$

- ▶ Since $DF(f) \in B(B(X, Y), Y) = B(X, Y)^*$, by the Riesz representation theorem there exists $\partial F/\partial f(f) \in B(X, Y)$ such that:

$$DF(f)\eta = \langle \eta, \frac{\partial F}{\partial f}(f) \rangle_{X^*} = \int_X \frac{\partial F}{\partial f}(f)(x)\eta(x)dx$$

Aside: Interpretation of $DF(f)$

- ▶ The Frechet derivative can be thought of as a generalised version of the total differential.
- ▶ Consider the total differential in finite dimensions. Let $X \subset \mathbb{R}^n$, $Y = \mathbb{R}$ and $f : X \rightarrow Y$. Then, the total differential of f is given by:

$$df = \sum_{i=1}^n \frac{\partial f}{\partial x_i} dx_i$$

- ▶ This looks similar to the “total differential” of the infinite dimensional function $F : B(X, Y) \rightarrow Y$ that was just defined:

$$DF(f)\eta = \int_X \frac{\partial F}{\partial f}(f)(x)\eta(x)dx$$

Back

Recursive Representation (Appendix)

- Aggregate states: $\{z, g\}$; idiosyncratic states: $x = (a, y)$.
- Given a belief $dg_t(x) = \tilde{\mu}^g(z_t, g_t)dt + \tilde{\sigma}(z_t, g_t)dB_t^0$, household choose c to solve HJBE:

$$\begin{aligned} 0 = & \max_{c \in \mathcal{C}(x, z, g)} \left\{ \rho V(x, z, g) + u(c) + D_x V(x, z, g) \mu^x(c, x, z, Q(z, g)) \right. \\ & + \lambda(x) (V(x + \gamma^x(x), z, g) - V(x, z, g)) + \text{higher order terms} \\ & + \partial_z V(x, z, g) \mu^z(z) + 0.5 (\sigma^z(z))^2 \partial_{zz} V(x, z, g) \\ & \left. + \int_{\mathcal{X}} \left(\frac{\partial V}{\partial g}(x, z, g(y)) \tilde{\mu}^g(y, z, g) + \text{higher order terms} \right) dy \right\} \\ & \text{s.t. BCs } \Psi(V)(x) = 0, \Phi(V)(x) \geq 0 \end{aligned}$$

- For optimal policy rule, $c^*(x, z, g)$, for z_t , population density, g , evolves by KFE:

$$dg_t(x) = \mu^g(c^*(x, z, g), z_t, g_t)dt - \text{div}[\sigma^a g_t(x)]dB_t^0. \quad (2)$$

- In equilibrium $\tilde{\mu}_g = \mu_g$.

“Master Equation” Representation of Equilibrium

- ▶ Suppose law of motion is $dg_t(x) = \mu^g(z_t, g_t)dt + \sigma(z_t, g_t)dB_t^0$.
- ▶ “Master equation” substitutes KFE, market clearing, & belief consistency into HJBE.
- ▶ Equilibrium value function $V(a, n, z, g)$ characterized by one PDE:

$$\begin{aligned} 0 = & -\rho V(a, n, z, g) + u(c^*(a, n, z, g)) + \mathbf{1}_{a_t \leq \underline{a}} \psi(a_t) \\ & + \partial_a V(a, n, z, g)(w(z, g)n + r(z, g)a - c^*(a, n, z, g)) + \text{Higher order terms} \\ & + \lambda(x) (V(a, \check{n}, z, g) - V(a, n, z, g)) + \partial_z V(a, n, z, g) \mu^z(z) + 0.5 (\sigma^z)^2 \partial_{zz} V(a, n, z, g) \\ & + \int_{\mathcal{X}} \frac{\partial V}{\partial g}(y, z, g) (\mu^g(c^*(y, z, g), y, z, g) + \text{Higher order terms}) dy =: \mathcal{L}V \end{aligned}$$

where the optimal control c^* is characterised by:

$$u'(c^*(a, n, z, g)) = \partial_a V(a, n, z, g).$$